[Total: 8]

3 Fig 3.1 shows a displacement-distance graph for two sound waves, A and B, of the same frequency and amplitude at a particular instant. Wave A is travelling to the right and wave B is travelling to the left.

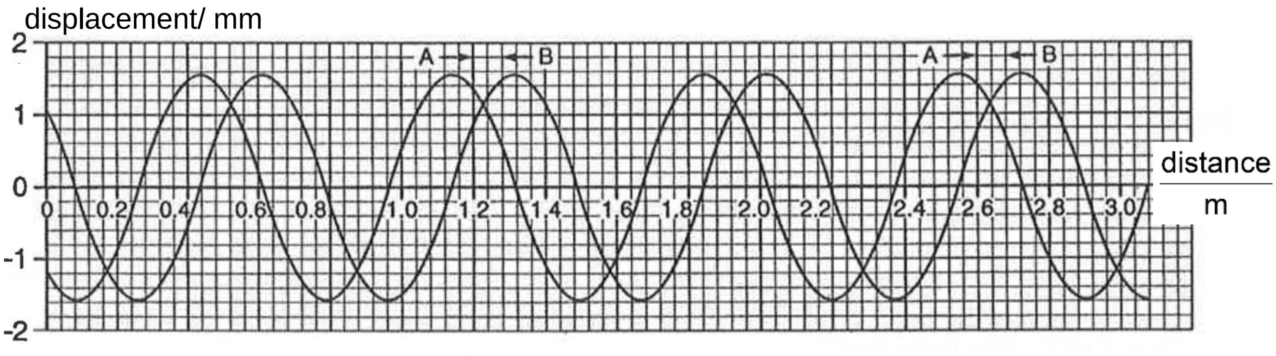

Fig. 3.1

(a) **(i)** Using Fig. 3.1, determine the wavelength of the two waves.

From x = 0.08 to 1.84, there are 2.5 wavelengths.

Wavelength = (1.84 – 0.08) / 2.5 = 0.704 m

wavelength = ... m [2]

(ii) The frequency of the sound is determined to be 469 Hz. Calculate the speed of sound to 4 significant figures.

$v = f\lambda$

$v = 469 \times 0.704 = 330.2$ m s^{-1}

[1]

(iii) The frequency and the wavelength of the sound were determined to a precision of ±5% and ±8% respectively. Write down the calculated value for the speed of sound in the form ($x \pm \Delta x$).

$\Delta v / v = \Delta f / f + \Delta\lambda / \lambda = 5 + 8 = 13\%$ M1

$\Delta v = \pm 40$ m s^{-1}

$v = (330 \pm 40)$ m s^{-1} A1

speed = (.................. ± ....) m s^{-1} [2]

(b) **(i)** State the phase difference between the two waves at the point where distance = 1.40 m in the instant shown in Fig. 3.1. Explain your answer.

π radians B1 (must have the unit)

Since the vector sum of the displacements of the two waves is zero, the point is a node. B1

....

....[2]

(ii) Hence calculate the maximum displacement of the resultant wave in the instant shown in Fig. 3.1. Explain your working clearly.

Antinode is ¼ of wavelength away.

At antinode, distance = 1.40 – 0.70/4 = 1.22 m B1

Displacement = 1.1 + 1.1 = 2.2 mm B1

maximum displacement = ...[2]

(iii) The maximum displacement of the resultant wave increases to a maximum value some time *t* later. Calculate the value of *t*.

From graph, distance between the two peaks = 0.16 m M1

Time for two peaks to meet, t = (0.16 / 2) / 330 = 0.24(2) ms A1

t = ...ms [2]

(iv) Deduce the maximum value of the maximum displacement in **(iii)**.

When the two waves meet in phase at their respective maximum displacements,

Maximum displacement of resultant wave = 1.6 + 1.6 = 3.2 mm

maximum displacement = ...mm [1]

[Total: 12]

4 (a) A circuit is set up as shown in Fig. 4.1. Cell A has an electromotive force (e.m.f.) of 2.0 V